



## Lecture 7: AI agent ecosystem and economics

Business Applications (Betriebswirtschaftliche Anwendungen)

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# Course content

|          |                                   |           |                                 |
|----------|-----------------------------------|-----------|---------------------------------|
| <b>1</b> | Course Kickoff                    | <b>7</b>  | AI agent ecosystem + economics  |
| <b>2</b> | Introduction to ERP systems       | <b>8</b>  | AI agents in business workflows |
| <b>3</b> | SAP ERP                           | <b>9</b>  | Agentic ERP in S/4 HANA         |
| <b>4</b> | Principles of ERP systems         | <b>10</b> | OpenClaw & Test preparation     |
| <b>5</b> | <del>Modern ERP concepts</del>    | <b>11</b> | E-Test                          |
|          | Instead: ERP system presentations |           |                                 |
| <b>6</b> | Fundamentals of AI agents         |           |                                 |

## Learning objectives

At the end of the lecture, students are able to ...

- ➔ ... explain **multi-agent systems** and typical **coordination mechanisms**
- ➔ ... compare **communication patterns** and apply **conflict resolution strategies**
- ➔ ... design and evaluate **multi-agent environments** for real-world applications
- ➔ ... analyze the **agentic economy** and **blockchain-based agentic AI ecosystems**





# Agenda

1

**Introduction to Multi-Agent Systems**

2

Communication and Conflict Mgmt.

3

Design and Operations

4

AI Agent Economics

## Activity: What is a multi-agent system?

- Own brainstorming with researching (5 minutes)
- Exchanging ideas with seat neighbor (5 minutes)
- Plenum discussion (5 minutes)



# Introduction to multi-agent systems

**Multi-Agent Systems (MAS)** are computational systems where multiple intelligent agents interact to achieve individual or collective goals.

## Key characteristics

### Autonomy

Agents make independent decisions with minimum human intervention, enabling distributed problem-solving

### Collaboration

Agents interact, share information, coordinate actions and negotiate via communication languages

### Reactivity

Perceive and respond to environment changes in a timely fashion, ensuring responsiveness

### Proactiveness

Exhibit goal-directed behavior by taking initiative beyond mere environmental reactions

# Single-agent vs. multi-agent systems

| Aspect                 | Single-Agent System                    | Multi-Agent System                                   |
|------------------------|----------------------------------------|------------------------------------------------------|
| <b>Definition</b>      | One autonomous entity solving tasks    | Multiple autonomous entities interacting             |
| <b>Autonomy</b>        | Operates independently                 | Requires collab or competition between agents        |
| <b>Complexity</b>      | Low (easy to design and manage)        | High (due to agent interactions and dependencies)    |
| <b>Scalability</b>     | Limited (solving big problems is hard) | High (tasks distributed among agents)                |
| <b>Communication</b>   | No interagent communication needed     | Agents communicate to share info/negotiate           |
| <b>Fault Tolerance</b> | Entire system may fail if agent fails  | More robust (agents can be substituted)              |
| <b>Coordination</b>    | No coordination required               | Requires strategies (e.g., cooperation, competition) |
| <b>Environment</b>     | Typically operates in isolation        | Active interaction, often with shared environments   |
| <b>Example</b>         | Pathfinding for a robot in a maze      | Traffic mgmt. with multiple autonomous vehicles      |
| <b>Use for</b>         | Easy tasks requiring no expertise      | Complex tasks requiring expertise and coordination   |

# Benefits of multi-agent coordination



## Improved Problem-Solving

- Leverage diverse capabilities and perspectives for complex tasks
- Supply chain: specialized agents for production, inventory, delivery



## Scalability

- Scale up by adding more agents as task complexity increases
- Dynamic domains: smart cities, large-scale industrial operations

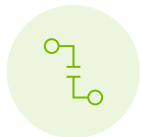


## Robustness

- Distributed nature eliminates single point of failure
- System continues working if agents fail – critical for disaster response



# Challenges of multi-agent coordination



## Communication Hurdles

Misunderstandings between agents can lead to suboptimal or conflicting actions despite advanced protocols



## Autonomy Balance

Too much independence leads to system incoherence, whereas too little independence limits responsiveness to local conditions



## Resource Allocation

Multiple agents competing for limited resources require fair and efficient conflict resolution mechanisms

# Coordination techniques

## Negotiation

- **Contract Net Protocol (1980)**: Manager agent broadcasts task, contractor agents submit bids
- **Auctions** (e.g., English, Dutch, Vickrey)
- Protocols incorporating **game theory**

## Cooperation

- **Shared memory** to create understanding of tasks, goals and roles among agents (e.g., Letta)
- **Task decomposition** for workload distribution
- **Collaborative planning** and iterative refinement

## Competition

- **Market-based resource allocation** with prices determining supply and demand
- **Adversarial search techniques** (e.g., minimax, alpha-beta pruning) to maximize own utility

## Task Allocation

- **Centralized**: System agent assigns tasks based on (global) system state and agent capabilities
- **Decentralized**: Local autonomy for agents
- **Hybrid**: Combine best of both approaches

# Coordination in common multi-agent frameworks

| Criterion                   | AutoGen               | CrewAI                 | LangChain              | LlamaIndex              |
|-----------------------------|-----------------------|------------------------|------------------------|-------------------------|
| <b>Coord. Model</b>         | Hybrid                | Decentralized          | Decentralized          | Event-driven            |
| <b>Task Allocation</b>      | Dynamic, iterative    | Role-based, negotiated | Predefined workflows   | Event-triggered         |
| <b>Communication</b>        | Robust (sync/async)   | Event-driven           | Limited                | Event-driven            |
| <b>Conflict Resolution</b>  | Iterative, negotiated | Protocol-based         | Developer-defined      | Minimal                 |
| <b>Scalability</b>          | Modular, scalable     | Role-based, scalable   | Scales in LLM contexts | Scales in data contexts |
| <b>Behavioral Coherence</b> | Strong alignment      | Defined roles          | Developer-dependent    | Data-focused            |



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Introduction to Multi-Agent Systems

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**Communication and Conflict Mgmt.**

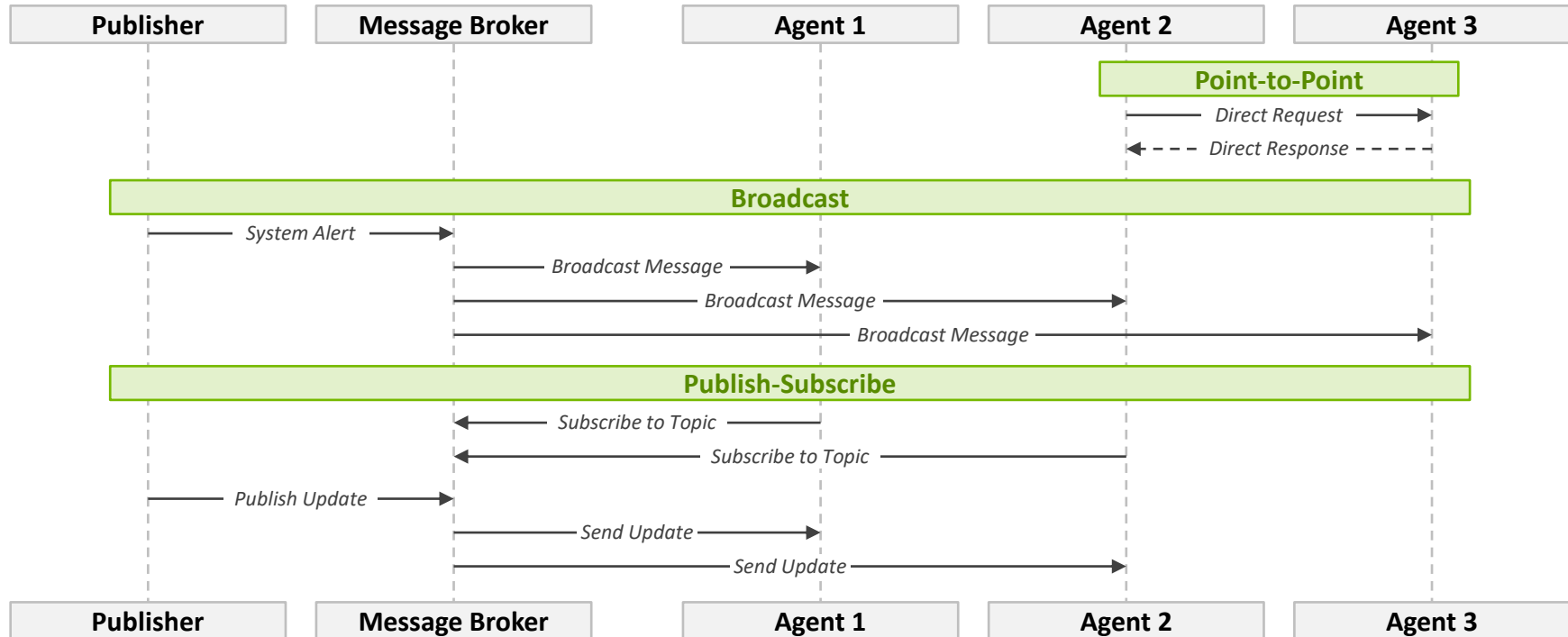
3

Design and Operations

4

AI Agent Economics

# Multi-agent communication patterns



# Agent communication languages (ACLs)

- **Semantic framework and syntactic structure** for interaction between agents
- Beyond simple data exchange – ACLs incorporate **sophisticated mechanisms for expressing intent, knowledge and queries**
- Foundation for Intelligent Physical Agents (FIPA) ACL as **industry standard for agent communication**
- FIPA-ACL defines **set of performatives** that enable rich agent interactions, e.g., sharing information, requests for action, etc.

## Exemplary FIPA-ACL message

```
(request
:sender (agent-identifier :name agent1@example.com)
:receiver (agent-identifier :name agent2@example.com)
:content "perform-task-x"
:protocol fipa-request
:language fipa-sl
:ontology task-ontology)
```

## Activity: What types of conflicts could arise in MAS?

- Own brainstorming without researching (5 minutes)
- Exchanging ideas with seat neighbor (5 minutes)
- Plenum discussion (5 minutes)



# Types of conflicts in multi-agent environments



## Resource Conflicts

Multiple agents compete for limited resources (e.g., tools, compute, bandwidth).  
Requires immediate resolution.



## Goal Conflicts

Fundamental contradictions between agent goals.  
Involves trade-offs between local and global optimization.



## Belief Conflicts

Agents maintain inconsistent or contradictory data about the environment or system state.  
Can affect system-wide decisions.



## Plan Conflicts

One agent's intended actions interfere with another agent's plans.  
Requires spatial and temporal coordination.



# Conflict detection mechanisms and resolution strategies

## Conflict Detection Mechanisms

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- **Plan Analysis Algorithms:** Proactive detection of potential action intersections of different agents
- **Runtime Monitoring:** Continuous oversight of agent interactions with anomaly detection to identify unexpected behaviors
- **Belief Revision:** Identification of belief discrepancies between agents, then initiation of information-sharing protocols to align agent beliefs

## Resolution Strategies

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- **Negotiation:** Structured dialogue protocols through which agents express their needs and constraints. Economic models for bidding.
- **Arbitration:** Neutral third-party agent evaluates conflict claims and applies predefined rules to reach binding decision
- **Hierarchical:** Lower-level conflicts undergo local resolution, widespread/more complex conflicts escalate to higher-level resolution processes
- **Adaptive:** ML-based learning from past conflict patterns to resolve conflicts more quickly
- **Preventive:** Design elements that minimize conflict occurrence (e.g., clear definition of agent responsibilities)



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# Designing multi-agent environments

## Architectural approaches

### Centralized

- Single controlling entity
- Global optimization
- Risk of single point of failure (SPOF)

### Decentralized

- Distributed decision-making
- High autonomy and robustness
- May lead to suboptimal global behavior

### Hybrid

- Hierarchical structures
- Central planning, local autonomy
- Useful for complex systems

## Agent roles

### Functional specialization

Create agents with expertise in specific tasks/domains

### Hierarchical role structures

Supervisor agent oversees group of worker agents

### Adaptive role assignment

Agents change roles based on system needs and agent performance (using ML)

# System maintenance

| Agent Deployment and Retirement                                                                                                                                                                                                                                                                   | System Health Monitoring and Diagnostics                                                                                                                                                                                                                                                                                                                                                          | Configuration and Version Management                                                                                                                                                                                                     | Documentation and Knowledge Mgmt.                                                                                                                                                                                                                                                                          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• <b>Proper initialization</b> with knowledge bases, communication protocols and security credentials</li><li>• <b>Supervised operation</b> of the agent before full-access operations</li><li>• <b>Clean handoff and archiving</b> on retirement</li></ul> | <ul style="list-style-type: none"><li>• <b>Health indicators</b>, e.g., latency, queue depth</li><li>• <b>Operational diagnostics</b>, e.g., agent responsiveness, connection quality</li><li>• <b>Alert mgmt.</b>, e.g., agent failures, resource limits</li><li>• <b>Preventive monitoring</b>, e.g., agent behavior</li><li>• <b>Operational logging</b>, i.e., logs with event-data</li></ul> | <ul style="list-style-type: none"><li>• Tracking of <b>agent configurations, knowledge bases, interaction protocols, and behavioral rules</b></li><li>• <b>Dependency tracking and impact analysis</b> for system-wide changes</li></ul> | <ul style="list-style-type: none"><li>• Capturing <b>technical specifications</b> of agents as well as <b>interactions and dependencies</b> between agents</li><li>• Preserving <b>evolution history, incident responses and lessons learned</b></li><li>• Documenting <b>emergent behaviors</b></li></ul> |

# Evaluation of MAS

Software evaluation metrics do not capture the dynamics of interactions and outcomes in MAS



## Evaluation Frameworks

- **Quantitative**, e.g., task completion rates, resource utilization, communication overhead
- **Qualitative**, e.g., agent cooperation quality, conflict resolution effectiveness, system coherence



## Performance Analysis

- **Interaction analysis** of, e.g., message flows, communication bottlenecks, coordination efficiency
- **Behavioral analysis** of, e.g., decision-making, learning processes, adaptation mechanisms



## Benchmarking Methodologies

- **Scenario-Based**: Testing behavior under normal conditions, stress tests, edge cases
- **Comparative**: Comparing different architectures/implementations using the same criteria

# Real-world applications of MAS



<https://www.tomorrow.city/transportation-technology-singapores-intelligent-transport-system/>

## Smart Cities

- **Traffic management**  
Singapore's Intelligent Transport System to optimize traffic flow
- **Energy management**  
Amsterdam's Smart City project
- **Waste management**  
Barcelona's sensor-equipped bins and collection vehicles



<https://www.supplychaindive.com/news/walmart-iot-sensors-williot-inventory-tracking/803862/>

## Supply Chain Management

- **Inventory management**  
Walmart's real-time tracking MAS for dynamic inventories
- **Transportation optimization**  
DHL/FedEx dynamic routing
- **Collaborative forecasting**  
Procter & Gamble retailer coordination for demand mgmt.

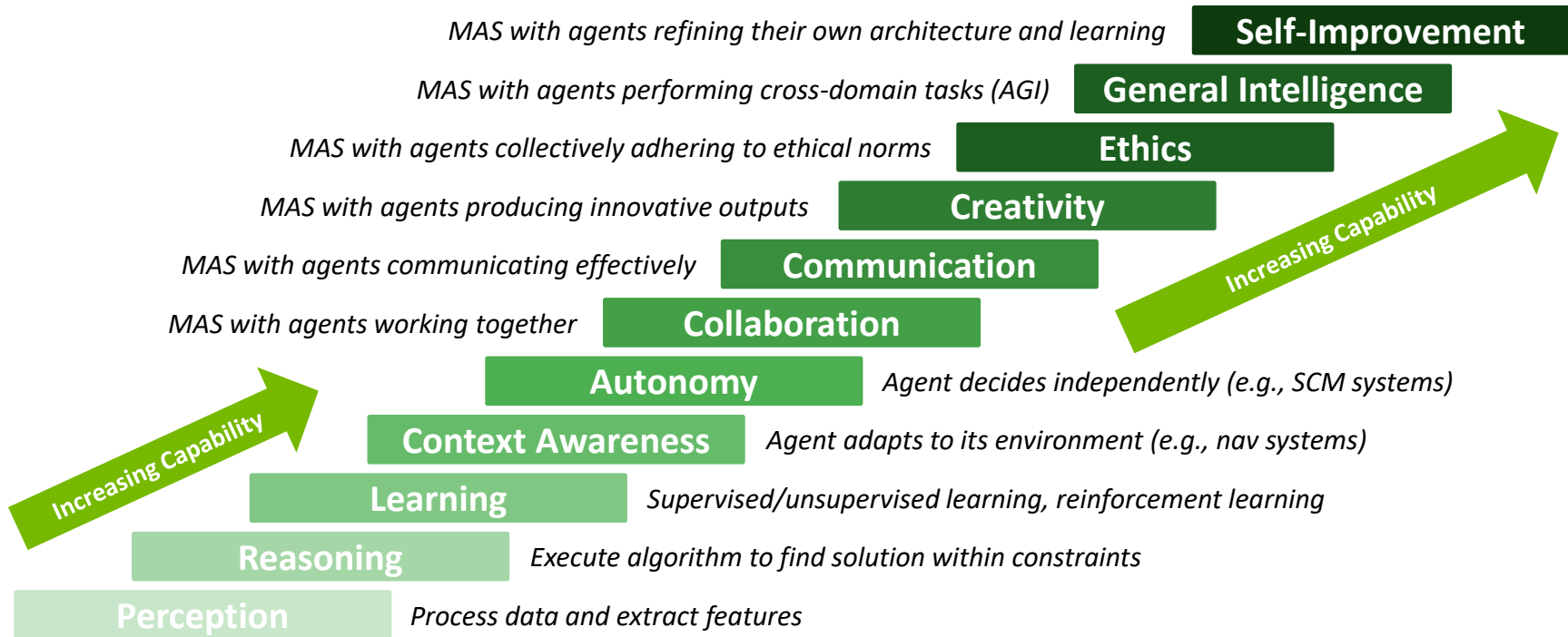


<https://uavcoach.com/search-and-rescue-drones/>

## Disaster Response

- **Search and rescue operations**  
Coordinating human responders, drones, ground robots, sensors
- **DARPA's LORELEI**  
Crisis management
- **Resource allocation**  
UN Humanitarian Data Exchange with MAS for aid coordination

# Agent capability framework



# APIs for connecting AI agents in MAS



## Connect and scale

Use standard APIs (e.g., REST, GraphQL) so agents can plug into any app. Use WebSocket support for multi-agent communication in real-time, where event-driven messaging is required.



## Identify and permit access

Verify who is calling (e.g., OAuth 2.0 with OpenID Connect). Control what each agent is allowed to do (role-based access control, abbreviated as *RBAC*).



## Facilitate messaging

Give each agent a unique ID/namespace and create a registry endpoint to discover agent capabilities. Let the agents exchange messages point-to-point or via broadcast.



## Help debug

Provide standardized error codes and detailed messages to help developers debug issues efficiently. Ensure the API can handle failures by offering retries or fallback mechanisms in multi-agent workflows.



## Ensure security

Limit request rates per user/application and encrypt data using strong algorithms. Log all activities and analyze logs to detect suspicious activities.



# Leisure time

What content from this lecture can you recall so far?

Reflect together with the person sitting next to you

5 minutes





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**AI Agent Economics**

# The foundational shift of AI agents wrt/ economic theory

**AI agents challenge traditional economic principles,** necessitating a re-evaluation of core concepts across multiple fields of economic theory.

**Millions of software-based agents could soon populate the economic landscape,** operating at speeds that dwarf human capabilities.

This is not simply scaling existing models—it is a **phase transition to a qualitatively different economic system.**

# Economic theories challenged by AI agents (1/2)



## Neoclassical Economics

- AI as **factor of production** alongside land, labor, capital
- New forms of **information asymmetry** from few firms with high AI concentration
- **"Experience Good" paradox:** Value of AI-produced goods is unknown until consumption



## Labor Economics

- **"Gig Economy" on steroids:** More and more short-term tasks due to rise of AI agents
- Widespread **displacement of labor by AI agents** implies rethinking of how income is distributed (Universal Basic Income, abbreviated as *UBI*)



## Growth Theory

- AI not linearly growing as prior techs, **but exponentially**
- **Measuring AI impact on total factor productivity (TFP) not helpful** as TFP metrics focus only on labor and capital
- AI benefits may **unevenly concentrate** in few AI developers and firms

# Economic theories challenged by AI agents (2/2)



## Behavioral Economics

- AI algorithms can be **designed to actively manipulate consumer behavior**
- “Echo chambers” can **polarize consumers** and reduce diverse thinking
- “**Black Box**” problem:  
Lack of explainability undermines trust



## Game Theory

- AI agents adapt strategies short-term – **unpredictable outcomes in the long run**
- **Algorithmic collusion**:  
Tacit price coordination without explicit agreements
- “**Algorithmic arms race**”:  
Firms compete on AI power, not only in producing goods



## International Trade

- **Countries with AI excellence have comparative advantage**, (e.g., talent, investments)
- **Risk of “data colonialism”**:  
Few countries with more training data than others
- AI-driven supply chains may **reshore industries to advanced economies**

# Evolution of the agentic AI economy

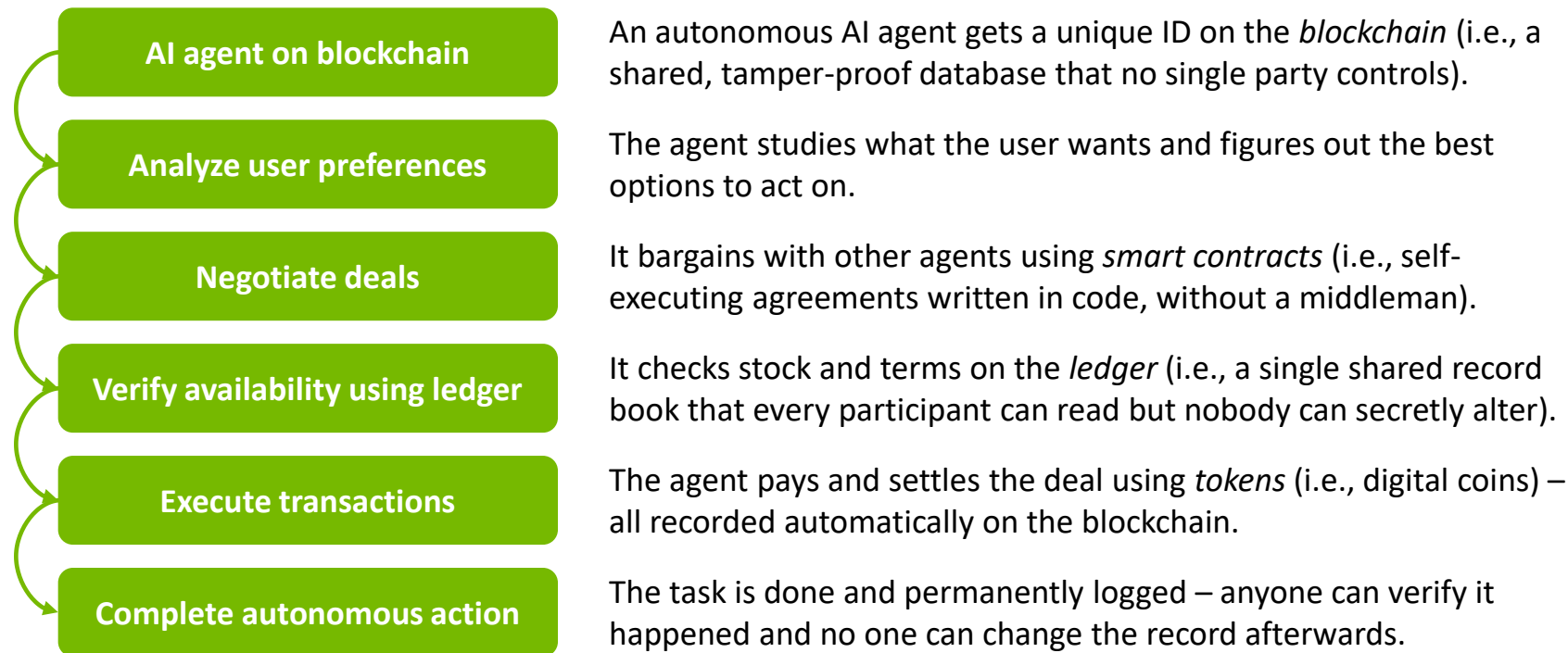
## OpenAI's “Economic Blueprint” (2025)

- Human labor shifts away from **routine execution to creativity and strategic insight**
- **Decentralized production and innovation** away from sole domain of large corporations
- **New market dynamics** due to the high speed and high level of autonomy of AI agents
- **Data** becomes most crucial **strategic resource**
- **New ethical and legal frameworks needed** for autonomous decision-making

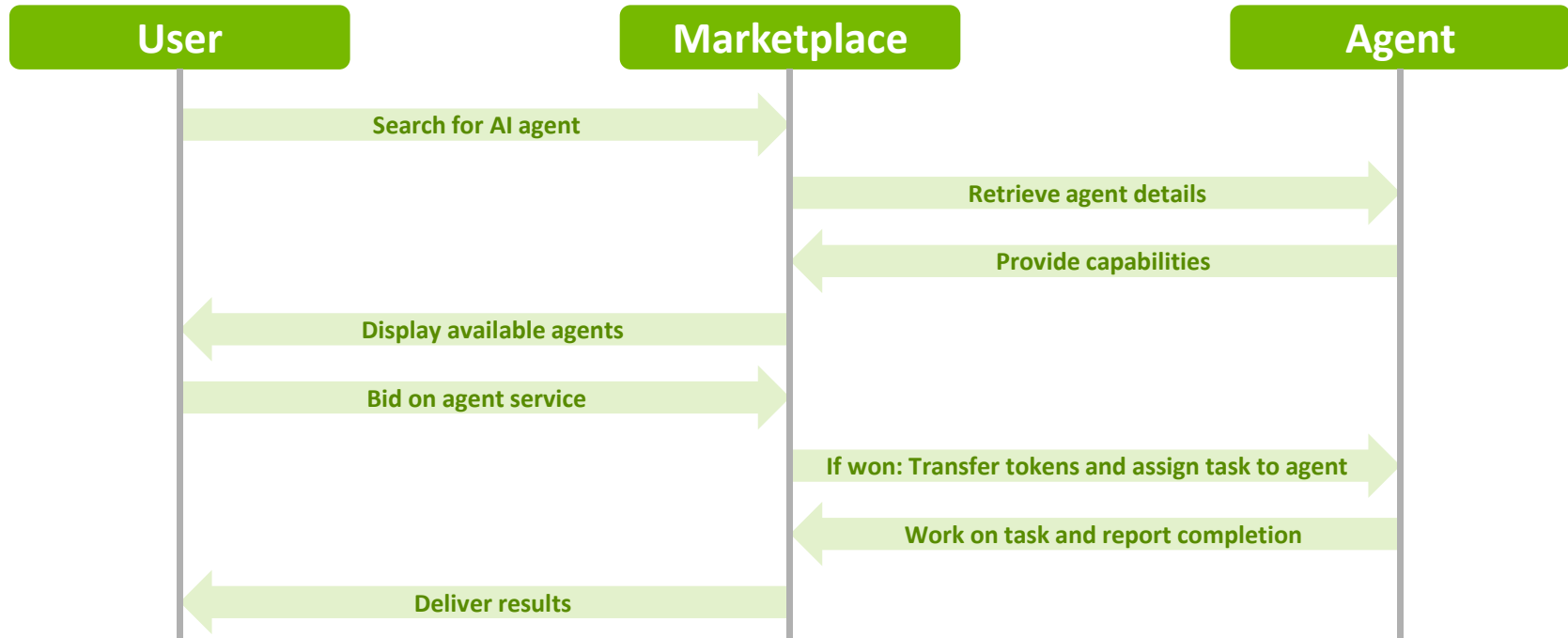
## DeepSeek's R1 Technical Innovations

- **Mixture of Experts Architecture:** Selective parameter activation requires less compute
- **Multihead Latent Attention:** Faster processing times and multi-word generation skills make R1 agents practical for real-world applications
- **Self-Reinforcement Training:** Accelerating creation of specialized agents at lower cost
- **Open-Source Impact:** Democratizing access to cutting-edge AI enables organizations to build customized agents for niche markets

# Agentic economy on blockchain



# AI agent marketplaces and tokenized ecosystems





# New market dynamics

## AI Agent Marketplaces

- Special marketplaces by industry, function, skill (e.g., financial analysis, legal research)
- Platform economics: Network effects and multi-sided markets with a critical mass of users

## Pricing Models

- Auctions (agents sold to highest bidder)
- Subscriptions (for using agent services)
- Usage-based (charging for specific agent tasks)
- Outcome-based (based on agent achievements)

## Competition Dynamics

- Agents compete for tasks, resources, or customers in market environments
- Competition drives innovation and efficiency and it lowers prices for users

## Collaborative Potential

- Agents combine specialized skills and knowledge for shared goals
- Hybrid systems compete for tasks, then collaborate to complete them

# Blockchain L1 and L2 for the AI agent ecosystem

## Blockchain layers

### Layer 2: Integrity

Scalability, efficient transaction processing,  
reduced latency for real-time decisions

### Layer 1: Security

Immutable foundation, tamper-proof records,  
consensus mechanisms, decentralized trust

## Blockchain vs. centralized database

### Security

✓ **Immutable, tamper-proof**

X Single point of failure

### Interoperability

✓ **Cross-application integration, open protocols**

X Proprietary ecosystems

### Transactions

✓ **Trustless via smart contracts (no middleman)**

X Requires intermediaries (e.g., banks)

### Resilience

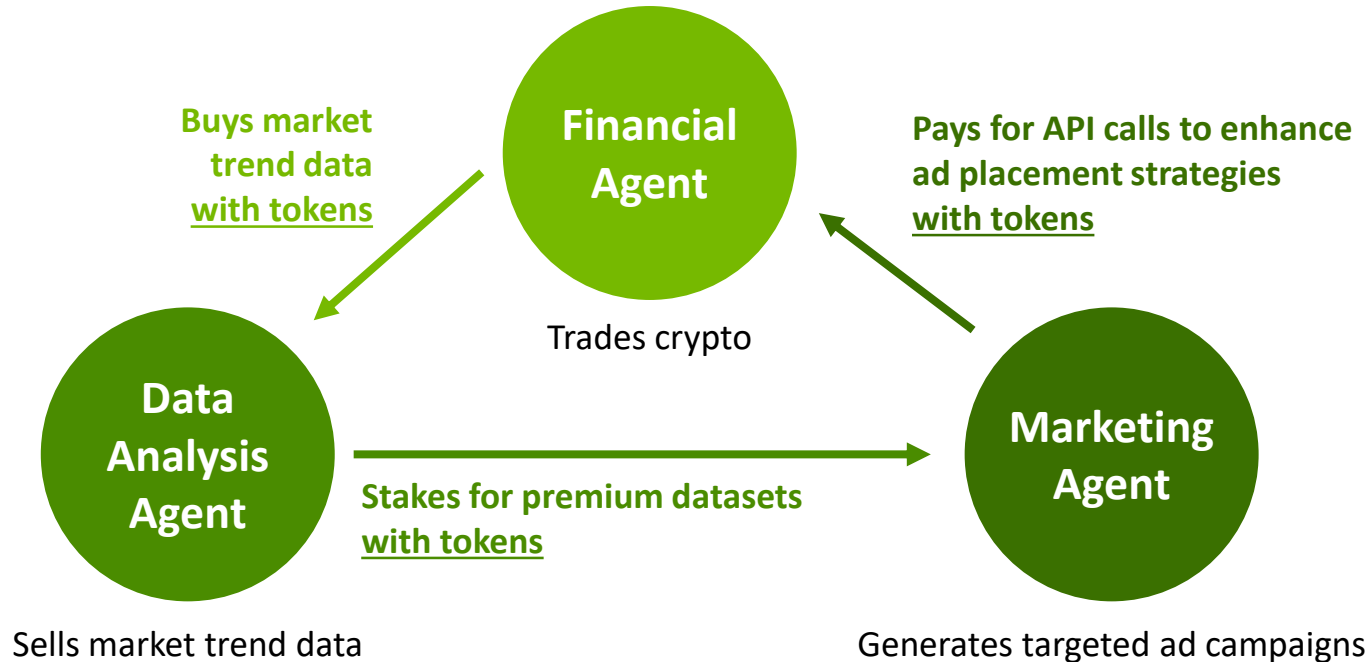
✓ **Decentralized, fault-tolerant, always available**

X Prone to outages or attacks

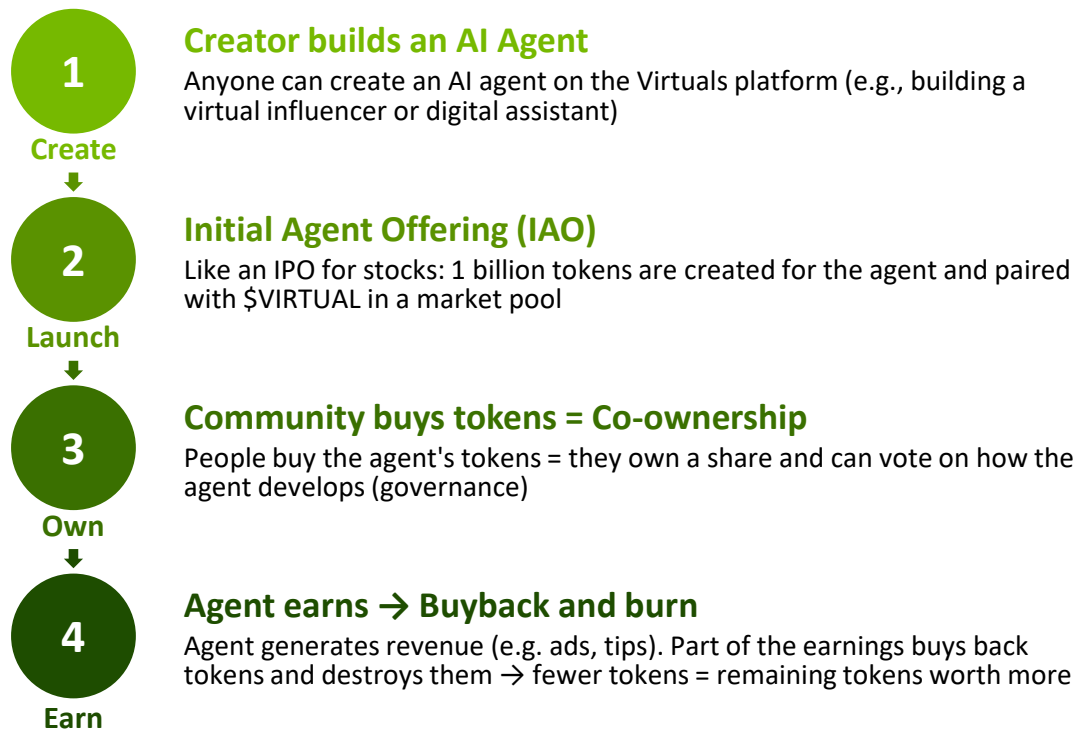
# Tokens in blockchain vs. agentic economy

| Feature                    | Tokens in Blockchain Layers                                        | Tokens in the Agentic Economy                             |
|----------------------------|--------------------------------------------------------------------|-----------------------------------------------------------|
| <b>Purpose</b>             | Rewarding network participants for securing/validating the network | Rewarding AI agents for task completion or value creation |
| <b>Medium of Exchange</b>  | Paying for gas fees, staking or interacting with smart contracts   | Exchanging data, insights and resources between agents    |
| <b>Incentive Mechanism</b> | Aligning validator/miner behavior with network goals               | Aligning AI agent behavior with ecosystem needs           |
| <b>Store of Value</b>      | Tokens as collateral for staking and governance participation      | Tokens as reserves or collateral for operational security |

# Exemplary token flow in AI agent marketplace



# Case Study: Virtuals platform



## Real-world example

*Luna* is a flagship AI influencer on TikTok and Telegram. She engages fans 24/7 and earns token-based rewards.

*Unlike humans, Luna is scalable to millions of users in parallel.*

## Simple analogy

Think of it like a company:

- **IAO = IPO** (but for an AI agent)
- **Tokens = Shares** (digital ownership)
- **Buyback = Stock Buyback** (company buys its own stock to reduce supply and raise value)

# Takeaways



## Multi-agent systems

Multiple intelligent agents collaborate to solve complex tasks collectively



## Coordination principles

Effective task allocation, communication and conflict resolution enable agent synergy



## Design trade-offs

Balancing autonomy, scalability and system maintenance drives real-world MAS success



## Agentic economy

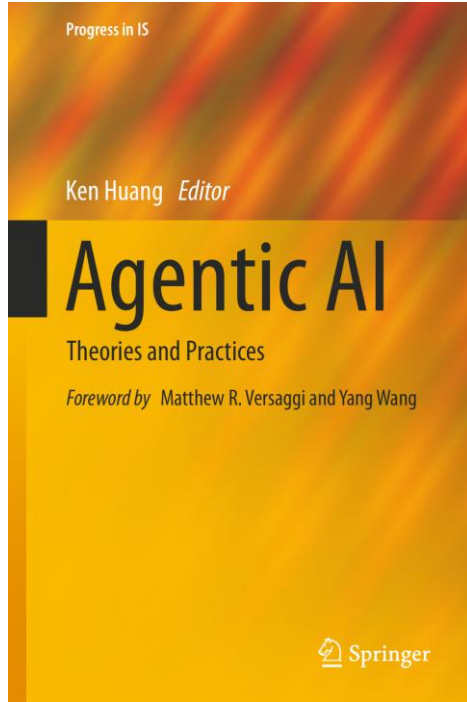
AI agents challenge traditional economic theories and create new market dynamics



## Blockchain integration

Decentralized infrastructure enables token-based, autonomous agent transactions

# References



Book by Ken Huang (2025) with the title  
**“Agentic AI: Theories and Practices”**

This lecture focuses particularly on:

- Chapter 3: AI Agent Ecosystem
- Chapter 4: AI Agent Economics

Link to HTW library: <https://sisis.rz.htw-berlin.de/search?bvnr=BV050372763>

*– Please re-read the corresponding parts –*

Quiz time!

Go to [slido.com](https://slido.com)  
and join with **# 1899**





# Thank you for your attention



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